

The Desk-Top Solar System

Big challenge · Indoors · Grown-up hands-on · Ages 7–11 · 45 min

Brought to you by Chike — “Hmmm, what should we create today?”

Build the planets on wire arms so you can spin them and see who gets round the sun first.

WHAT YOU NEED

- Thick cardboard for the base
- Clay or scrunched foil for planets
- Skewers, straws, or stiff wire
- Tape
- Markers

BEFORE YOU START

A grown-up handles skewers or wire and blunts every sharp end with tape or a cork.
Work on a hard surface, not your lap.

How to make it

- 1 Cut a circle of thick cardboard for the base and mark the sun in the center.
- 2 Draw four rings around the sun, each one further out than the last.
- 3 Roll four planets from clay: two small and rocky for the inner rings, two larger for the outer ones.
- 4 Push a taped skewer through each planet and anchor it to the base so it can swing around the sun.
- 5 Spin the inner planet all the way round once, then spin the outer one. Count how far each one travels.
- 6 Label each ring with how long that planet takes to go round once.

TALK ABOUT IT

If you lived on the outer planet, how old would you be in that planet's years?

TRY THIS TOO

Add a moon on a shorter arm that goes around one of your planets while the planet goes around the sun.

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Planets further from the sun take longer to go around it. The outer planets have a much bigger loop to travel, and they also move more slowly along it. That is why one trip around the sun takes Earth a year, and takes Neptune about 165 of our years.